



Fig. 3: HyperTerminal new connection



Fig. 4: COM port properties window

gain, programmable watchdog timer with internal oscillator, a serial peripheral interface (SPI) port, IEEE standard 1149.1 compliant JTAG test interface (also used for accessing the on-chip debug system and programming) and six software-selectable power-saving modes.

Circuit description

Fig. 2 shows the circuit for storing data on the AT24C64 PC EEPROM using an ATmega2560 AVR microcontroller. The circuit is powered by a standard 9V DC source. The 9V DC is converted into 5V DC using a 7805 regulator (IC1). The glowing of LED1 shows the presence of power in the circuit. Regulated 5V powers the circuit including the ATmega2560, MAX232, AT24C64 and ISP6 connector (CON2). CON3 is a 9-pin, D-type COM port connector used to interface with the PC for text

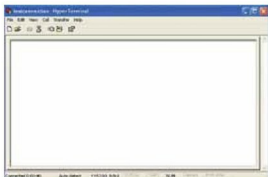


Fig. 5: HyperTerminal output window

file transfer to the AT24C64 through the microcontroller (IC2). Address pins A0, A1 and A2 of AT24C64 are connected to ground and SDA and SCL pins are connected to pins 44 and 43 (PD1 and PD0) of the microcontroller, respectively. Resistors R5 and R4 (4.7K each) are external components connected to SDA and SCL pins of AT24C64, respectively. Pin 8 of AT24C64 is connected to +5V, while its pin 7 (WP) is not used here.

ATmega2560 (IC2) is the main controller in this circuit. A 16MHz crystal oscillator is used to generate the clock pulse. Communication between the microcontroller and the PC is done through MAX232 driver (IC3). Pins 2 and 3 of IC2 are connected to pins 12 and 11 of IC3, respectively. Pins 20, 21 and 22 of IC2 are serial data communication lines connected to ISP6 connector CON2. ISP6 connector is used in programming the microcontroller using STK500 board.

Whenever the program doesn't function properly, you can reset the microcontroller by momentarily pressing reset switch S1.

Software program

The software for the microcontroller is written in 'C' language using the IAR Embedded Workbench integrated development environment. IAR Embedded Workbench is being developed by IAR Systems and ATMEL developers in parallel and hence it generates an optimised code which uses full 'C' coding capabilities of AVR devices. AVR development tools for embedded sys-

tems can be downloaded for free from IAR website www.iar.com. For details of IAR Embedded Workbench, you may refer to 'A Beginners' Guide to ATMEL AVR Development' article published in January issue of EFY.

To start the application, click 'IAR Embedded Workbench' icon in 'All Programs' menu of Windows. From the main menu, select

Project→Create New Project and start an AVR Studio compatible C project. Generate the Intel hex file for burning the code into ATmega2560 microcontroller as follows: Project→Options→Linker (under Category)→Extra Output→Generate Extra Output (to click)→Override Default (to click)→Type project name with .hex extension→Output Format→Intel-standard (to select). Press 'ok' after all these settings are done.

The source code (I2C_EEPROM.c) to store data on an I²C EEPROM using an AVR microcontroller is given at the end of this article along with comments for instructions. It lets you store 4096 bytes of data in AT24C64 using principles of I²C communication.

Downloading the code into ATmega2560

Download the code into the AVR using appropriate tools. Two options to download the code are:

Using AVR Studio and STK500.

To program the ATmega2560, connect the 6-wire cable between ISP connectors of the STK500 board (marked as ISP6) and the target board (marked as ISP con). Connect a serial cable from 'RS232 CTRL' connector on the STK500 board to a COM port on the PC. Now start AVR Studio 4.0 without opening the project file.

From the AVR Studio output window, proceed as follows: Main Menu→Tools→Program AVR→Select AVR Programmer. Press 'Connect...' after selecting STK500 or AVRISP in